

Benchmark Minerals Intelligence — Demo Script

Showcasing Native Apps, Cross-Cloud Marketplace Deployment & AI Capabilities on Snowflake

Demo Overview

Benchmark Minerals Intelligence, a critical minerals pricing data provider, builds a complete data product on Snowflake and deploys it globally via the Snowflake Marketplace — all without moving data outside the platform. The entire project was built using **Cortex Code (CoCo)**.

Key Themes: - Building Native Apps with Streamlit, Semantic Views & Cortex AI - Publishing & deploying Marketplace listings cross-cloud (EU → US → Azure) - AI-powered analytics with Cortex Agent, Cortex Search & Cortex Analyst - Developer productivity with Cortex Code (CoCo) — the AI that built all of this

Accounts Used:

Account	Region	Role
IE_DEMO10 (Provider)	AWS EU-West-1 (Ireland)	ACCOUNTADMIN
COLM_USWEST (Consumer)	AWS US-West-2 (Oregon)	ACCOUNTADMIN
AZWEU (Consumer)	Azure West-EU	ACCOUNTADMIN

Demo Flow (~25 min)

[Diagram — see markdown version]

Time	Section	Focus
0–3 min	Snowsight Tour	Data catalog, BENCHMARKMINERALS_DB
3–8 min	AI Capabilities	Semantic Views, Cortex Search, Agent, Intelligence
8–12 min	Streamlit App	Dashboard, agent chat, live prices
12–16 min	Native App Build	Application package, manifest, declarative sharing
16–20 min	Cross-Cloud Deploy	Provider Studio → Oregon → Azure
20–25 min	Cortex Code	Live CoCo demo, how the project was built with AI

Act 1: The Provider Experience (IE_DEMO10)

1.1 Snowsight Overview

- Open **Snowsight** on IE_DEMO10
- Quick tour: Databases, Worksheets, Streamlit, Notebooks
- Show the **Catalog** — browse `BENCHMARKMINERALS_DB`

1.2 The Data

"Benchmark Minerals tracks daily pricing for critical minerals — lithium, cobalt, nickel, gold, platinum and more — across global exchanges."

Navigate to `BENCHMARKMINERALS_DB.PUBLIC` and show key tables:

Table	Description
<code>BATTERY_CRITICAL_MINERALS_PRICING</code>	Daily OHLC prices across materials & exchanges
<code>MARKET_INDICES</code>	Market index benchmarks
<code>BATTERY_SUPPLY_CHAIN</code>	Supply/demand forecasts joined with mining ops
<code>MINING_OPERATIONS</code>	75 mines with ESG risk scores
<code>BATTERY_CELL_MANUFACTURERS</code>	Cell manufacturing by country & chemistry
<code>ANALYST_REPORT_CHUNKS</code>	Parsed PDF analyst reports (731 chunks, 10 reports)

Diagram: - `BATTERY_CRITICAL_MINERALS_PRICING` - date `ASSESSMENT_DATE` - string `MATERIAL` - string `EXCHANGE` - float `PRICE_CLOSE_USD_PER_OZ` - `MARKET_INDICES` - date `INDEX_DATE` - string `INDEX_NAME` - float `INDEX_VALUE` - `MINING_OPERATIONS`

Act 2: AI Capabilities

2.1 Semantic Views — Natural Language to SQL

"We've created Semantic Views so anyone can query our data in plain English."

- Show `BATTERY_CRITICAL_MINERALS_PRICING_SV` and `MARKET_INDICES_SV`
- Open **Cortex Analyst** and ask:
- *"What is the latest gold price?"*
- *"Compare lithium and cobalt prices over the last 6 months"*
- *"Which metal had the biggest price increase this year?"*
- Show the generated SQL — Semantic Views map business terms to physical columns

2.2 Cortex Search — Unstructured Data

"We have 10 analyst reports from Goldman Sachs, World Gold Council, Sprott, WPIC and others"

— all indexed with Cortex Search."

- Show `ANALYST_REPORTS_SEARCH` service
- PDFs parsed with `PARSE_DOCUMENT`, chunked, and indexed
- Query: "What is Goldman Sachs' gold price forecast for 2026?"

2.3 Cortex Agent — The AI Analyst

"We've combined structured data (via Semantic Views) and unstructured data (via Cortex Search) into a single Cortex Agent."

[Diagram — see markdown version]

- Show `BENCHMARK_MINERALS_AGENT` in `AGENT_SCHEMA`
- Open **Snowflake Intelligence** and interact:
- "What are the top 5 most expensive metals right now?" → routes to Analyst
- "Summarise the key risks in the Sprott gold outlook" → routes to Search
- "Compare lithium and cobalt — price trend and analyst views?" → multi-tool

2.4 Streamlit App with Embedded Agent

"Our Streamlit app gives users an interactive dashboard AND an AI chat interface — all running natively in Snowflake."

Diagram: - Tab1Precious Metals / Pricing - Tab2Battery / Supply Chain - ChatAgent Chat / sidebar - LiveLive Price / Lookup

- Open `BENCHMARK_MINERALS_DASHBOARD`
- Walk through:
- **Precious Metals Pricing** — line charts, daily change heatmap, metric cards
- **Battery Supply Chain** — supply/demand forecasts, ESG scatter, manufacturing by country
- **"Let's Chat"** — sidebar agent chat via `CALL_ASK_BENCHMARK_AGENT`
- **Live Price Lookup** — real-time prices via External Access Integration
- Ask the agent a question live

Act 3: Building the Native App

3.1 Declarative Sharing

"We share data, AI, and app as a single product. Snowflake's declarative sharing makes this possible."

Diagram: - T8 Tables - SV2 Semantic Views - CS1 Cortex Search - AG1 Cortex Agent - NB1 Notebook - ST1 Streamlit App

- Show `BENCHMARK_MINERALS_NATIVE_APP_PKG`
- Walk through `manifest.yml`: ``yaml shared_content: databases:
 - `BENCHMARKMINERALS_DB`: schemas: - `AGENT_SCHEMA`: # Cortex Agent - `PUBLIC`: # Tables, Semantic Views, Cortex Search ``

3.2 Setup Script & Streamlit

- `setup.sql` — creates `CORE` schema with views over `SHARED_DATA`
- `streamlit_app.py` — consumer-facing dashboard, queries shared data through views

Act 4: Cross-Cloud Marketplace Deployment

"We publish once from Ireland — Snowflake handles replication everywhere."

[Diagram — see markdown version]

4.1 Publishing (IE_DEMO10)

- Open **Provider Studio**
- Show listing: `BENCHMARK_MINERALS_APP_LISTING`
- **Type**: External (Marketplace) · **Distribution**: Cross-region
- **Auto-fulfillment**: `SUB_DATABASE_WITH_REFERENCE_USAGE`
- Show **Provider Profile** with company logo

4.2 Consumer Install (COLM_USWEST — Oregon)

- Switch to **COLM_USWEST**
- **Marketplace** → Search "Benchmark Minerals" → **Get**
- Open installed app — Streamlit dashboard with live data from Ireland
- `SELECT * FROM`

`BENCHMARK_MINERALS_NATIVE_APP.PUBLIC.BATTERY_CRITICAL_MINERALS_PRICING LIMIT 10`

4.3 Data Listing (AZWEU — Azure)

- Switch to **AZWEU**
- **Catalog** → search for Benchmark Minerals data
- Open a **Notebook** to explore the shared data

Act 5: The Complete Picture

Diagram: - Tables8 Tables - SVs2 Semantic Views - SearchCortex Search - SLitStreamlit + Chat - CAgentCortex Agent · 3 tools - DB -.-> PKG - C1AppNative App + Streamlit - C1Data8 Tables referenced - C2ShareData Share - C2NBNotebook

Key Talking Points

Capability	What We Showed
Native Apps	Full app with Streamlit UI, setup scripts, shared data — single installable package
Declarative Sharing	Tables, Semantic Views, Cortex Search, Agent — all shared without copying data
Cross-Cloud	Published once from Ireland → auto-replicated to Oregon (AWS) and Azure
Cortex Analyst	Natural language → SQL via Semantic Views
Cortex Search	10 analyst PDFs parsed, chunked, and searchable
Cortex Agent	Multi-tool AI analyst: structured + unstructured data
Streamlit in Snowflake	Interactive dashboard with embedded AI chat
Snowflake Intelligence	End users query the agent directly from Snowsight
External Access	Live metal prices from external API
Cortex Code	AI pair-programmer that built the entire demo

Act 6: Built with Cortex Code (CoCo)

*"Everything you've seen today was built using **Cortex Code**, Snowflake's AI-powered development CLI."*

6.1 What is Cortex Code?

[Diagram — see markdown version]

- AI pair-programmer in your terminal — understands Snowflake natively
- Direct access to your account: queries, creates objects, deploys apps, manages listings
- **Built-in skills** for Streamlit, Native Apps, Cortex Agent, Semantic Views, Dynamic Tables, and more

6.2 What CoCo Built for This Demo

***Diagram:** - D18 Tables / realistic pricing data - D22 Semantic Views - D3Cortex Search / 10 PDFs parsed - D4Cortex Agent / 3 tools - D5Streamlit Dashboard / multi-tab + agent chat - D6Stored Procedure / CortexAgentService bridge - D7Native App Package / manifest + setup + UI - D8Marketplace Listing / cross-cloud replication*

What	How CoCo Helped
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Data generation	Realistic pricing data for 8 tables — minerals, exchanges, OHLC prices, ESG scores, manufacturers
Semantic Views	Created YAML definitions, validated, and deployed to Snowflake
Cortex Search	Parsed 10 PDFs with <code>PARSE_DOCUMENT</code> , chunked, created search service
Cortex Agent	Built 3-tool agent with orchestration instructions and sample questions
Streamlit app	Multi-tab dashboard — charts, heatmaps, agent chat, live prices
Stored procedure	<code>CortexAgentService</code> bridge (native SiS can't call Agent REST APIs)
Native App	<code>manifest.yml</code> , <code>setup.sql</code> , consumer Streamlit, full package
Marketplace	Published listing, cross-cloud auto-fulfillment, managed replication
Consumer app	Streamlit dashboard on <code>COLM_USWEST</code> pointing at installed native app
Debugging	Version/release channel issues, SV YAML errors, replication delays, encryption

6.3 Live CoCo Demo

"Let me show you CoCo in action."

- Open Cortex Code in the terminal
- Run live prompts:
- *"Show me the tables in `BENCHMARKMINERALS_DB`"* — instant SQL
- *"What's in the Cortex Agent?"* — describes agent config
- *"Add a new sample question to the Streamlit app"* — edits code directly
- *"What's the average gold price this month?"* — calls Cortex Analyst
- Key capabilities:
- **Multi-tool orchestration** — reads files, runs SQL, writes code, deploys in one conversation
- **Persistent memory** — remembers project context across sessions
- **Built-in skills** — specialised workflows for every Snowflake product
- **Iterative dev** — fix errors, re-deploy, validate without leaving the terminal

6.4 The Punchline

"This entire demo — from empty account to a published, cross-cloud, AI-powered data product on the Snowflake Marketplace — was built through conversations with Cortex Code. No copy-pasting from docs. No switching between tools. Just describe what you want, and CoCo builds it on Snowflake."

Appendix: Sample Questions for Live Demo

Structured Data (Cortex Analyst): 1. What is the latest gold price? 2. Compare lithium and cobalt

prices over the last 6 months 3. Which metal had the biggest price increase this year? 4. What are the top 5 most expensive metals right now? 5. Show me the price trend for platinum on LBMA

Unstructured Data (Cortex Search): 1. What is Goldman Sachs' gold price forecast for 2026? 2. What does the World Gold Council outlook say about gold demand? 3. What is the platinum market outlook from WPIC? 4. Summarise the key risks highlighted in the Sprott gold outlook 5. What are analysts predicting for precious metals prices this year?

Multi-Tool (Agent): 1. Compare lithium prices and what analysts are saying about the lithium market 2. How has nickel performed and what's the supply outlook?